10 — Roadmap and Milestones

Interpretable Machine Learning for Discovering Hidden Regulatory Switches in Non-Coding DNA

Field	Value
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Related docs	Charter, Modeling Roadmap, Risk Register

Roadmap Overview

```
Month:   1          2          3          4          5          6          7-9        10-12
        ██████████████████████████████████████████████████████████████████████████████
Phase 0: ████████
Phase 1:       ██████████████████
Phase 2:                        ████████████████████
Phase 3:                                              ████████████████
Phase 4:                                                                 ████████████
Writing:                   ████████████████████████████████████████████████████████

████ = Active development
████ = Writing / documentation (parallel track)
```

Month 1: Foundation

Objective

Establish data pipeline, environment, and first baseline model.

Milestones

#	Milestone	Target date	Success criterion
M1.1	**Project setup complete**	Week 1	Git repo initialized; directory structure created; requirements.txt defined; docs pack reviewed by team
M1.2	**Literature survey complete**	Week 2	≥ 20 papers read and annotated; annotated bibliography committed

M1.3	**Data pipeline functional**	Week 3	ENCODE cCREs downloaded for K562; sequences extracted from hg38; negative set generated; chromosome-holdout split defined
M1.4	**First baseline model trained**	Week 4	Logistic regression on k=4 k-mers; AUROC computed on test set; results logged

Deliverables

- [ ] Repository with directory structure and documentation
- [ ] Annotated bibliography
- [ ] Data pipeline: download → parse → extract → split
- [ ] EDA notebook with data visualizations
- [ ] First experiment log (`exp_001_logistic_kmer4`)
- [ ] Experiment registry initialized

Definition of Success at Month 1

> A reproducible pipeline that goes from raw ENCODE annotations to a trained classifier with logged metrics. At least one model produces AUROC > 0.60 (proving the task is learnable).

Month 3: Interpretable Baseline Complete

Objective

Complete Phase 1 (classical ML) with full interpretability analysis and robust validation.

Milestones

#	Milestone	Target date	Success criterion
M3.1	**Multi-model comparison**	Month 2, Week 2	LogReg, RF, XGBoost trained and compared on same split; comparison table produced
M3.2	**Feature importance analysis**	Month 2, Week 3	Top-20 features extracted per model; cross-referenced with JASPAR; motif overlap table generated
M3.3	**SHAP analysis complete**	Month 2, Week 4	SHAP summary plot for XGBoost; SHAP force plots for 5 sample regions
M3.4	**Negative set sensitivity analysis**	Month 3, Week 1	Experiment E7 complete; AUROC stable ($\Delta < 0.05$) across ≥ 2 negative sampling strategies

M3.5	**k-mer resolution analysis**	Month 3, Week 1	Experiment E3 complete; optimal k value identified
M3.6	**Feature ablation study**	Month 3, Week 2	Experiment E2 complete; contribution of each feature group quantified
M3.7	**Decision Gate 1→2 passed**	Month 3, Week 3	AUROC ≥ 0.75; interpretability verified; all gate criteria met
M3.8	**Phase 1 report written**	Month 3, Week 4	Internal report summarizing baseline results, interpretability findings, data quality assessment

Deliverables

- [ ] ≥ 5 completed experiments with full logging
- [ ] Model comparison table (accuracy + interpretability)
- [ ] Motif cross-reference report
- [ ] SHAP analysis artifacts
- [ ] Feature ablation results
- [ ] Negative set sensitivity analysis
- [ ] Internal Phase 1 report
- [ ] Kaggle notebook reproducing key results
- [ ] Updated experiment registry

Definition of Success at Month 3

> At least one classical model achieves AUROC ≥ 0.75. Feature importance analysis reveals biologically plausible patterns (≥ 30% of top-20 features match known motifs). All results are reproducible. Decision Gate 1→2 is passed.

Month 6: Deep Learning and Cross-Cell-Type Analysis

Objective

Complete Phase 2 (CNN) and cross-cell-type experiments. Begin research writing.

Milestones

#	Milestone	Target date	Success criterion
M6.1	**Shallow CNN trained**	Month 4, Week 2	CNN achieves AUROC ≥ 0.80 on test set; training completes on Kaggle GPU
M6.2	**CNN filter analysis**	Month 4, Week 4	First-layer filters visualized as sequence logos; ≥ 3 filters match JASPAR motifs

M6.3	**Saliency / Integrated Gradients**	Month 5, Week 2	Saliency maps generated for 100+ test regions; qualitative analysis documented
M6.4	**Cross-cell-type experiment**	Month 5, Week 4	Experiment E4 complete for K562 + GM12878; cell-type-specific motifs identified
M6.5	**Element-type classification**	Month 5, Week 4	Experiment E5 complete; enhancer vs. promoter classifier evaluated
M6.6	**Accuracy vs. interpretability report**	Month 6, Week 1	Trade-off analysis comparing all model families documented
M6.7	**Decision Gate 2→3 passed**	Month 6, Week 2	All gate criteria met; team decides whether to pursue pretrained embeddings
M6.8	**Draft research manuscript (introduction + methods)**	Month 6, Week 4	Introduction and methods sections drafted

Deliverables

- [ ] Trained CNN model with filter analysis
- [ ] Saliency map gallery (representative examples)
- [ ] Cross-cell-type comparison report
- [ ] Enhancer vs. promoter classification results
- [ ] Full model comparison matrix (classical + CNN)
- [ ] Accuracy-vs-interpretability analysis document
- [ ] Draft manuscript (intro + methods)
- [ ] Updated Kaggle notebooks

Definition of Success at Month 6

> CNN achieves AUROC $\geq$ 0.80. Cross-cell-type analysis reveals cell-type-specific regulatory features. At least one novel (non-JASPAR) candidate motif pattern is identified by the CNN. Research manuscript is in progress.

Month 12 Vision: Research-Ready Platform

Objective

Complete remaining phases (as time allows), submit or post publication, and ensure the project is documented for long-term continuity.

Milestones (Months 7–12)

#	Milestone	Target window	Success criterion

M12.1	**Pretrained embedding experiments** (Phase 3)	Months 7–9	Embeddings extracted from ≥ 1 pretrained model; comparison to baselines documented
M12.2	**Hybrid model** (k-mer + embedding)	Month 8–9	Hybrid classifier evaluated; SHAP on hybrid features
M12.3	**Full results manuscript**	Month 9–10	Complete draft including results, discussion, limitations
M12.4	**Model card(s)**	Month 10	Model card for best model documenting performance, limitations, intended use
M12.5	**Public code release**	Month 10	Cleaned repository with README, license, installation instructions
M12.6	**Publication submission**	Month 11	Submitted to target venue (workshop or journal)
M12.7	**Demo or visualization tool** (optional)	Month 12	Gradio app or static HTML showing interactive regulatory predictions
M12.8	**Project handoff documentation**	Month 12	Updated all docs; contributor onboarding verified with a new team member

Deliverables (Months 7–12)

- [ ] Pretrained embedding comparison report
- [ ] Complete research manuscript
- [ ] Model card(s)
- [ ] Public repository release
- [ ] Submitted paper or preprint
- [ ] (Optional) Interactive demo
- [ ] Full project documentation update

Definition of Success at Month 12

> The project has produced: 1. A published or submitted research paper on interpretable regulatory prediction. 2. A reproducible codebase usable by other researchers. 3. Documented evidence of regulatory switch-like sequence patterns. 4. A platform that can be extended by future contributors.

What "Success" Means at Each Stage

Stage	Success definition	Minimum acceptable outcome
Month 1	Pipeline works; first model trained	Data loads correctly; any AUROC > 0.5

Month 3	Interpretable baseline with motif validation	AUROC ≥ 0.75; ≥ 3 known motifs in top-20 features
Month 6	CNN + cross-cell-type analysis	AUROC ≥ 0.80; cell-type-specific patterns documented
Month 12	Publication + reproducible platform	Paper submitted; code public; handoff-ready

Minimum Viable Research Contribution (MVRC)

Even if the project stops at Month 3, the following constitutes a valid research contribution:

1. Reproducible pipeline for ENCODE-based regulatory classification.
2. Multi-model comparison showing accuracy-interpretability trade-offs.
3. Feature importance analysis connecting k-mer/motif features to known biology.
4. Honest documentation of what works and what doesn't.

This can be published as a workshop paper or technical report.

Milestone Dependency Graph

```
M1.1 (Setup)
 ■
 ■■■ M1.2 (Literature) ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
 ■                                                                ■
 ■■■ M1.3 (Data pipeline) ■■→ M1.4 (First baseline)              ■
                                  ■                               ■
                         M3.1 (Multi-model) ■■→ M3.5 (k sweep)   ■
                          ■                     ■                 ■
                         M3.2 (Importance) ■■→ M3.6 (Ablation)   ■
                          ■                                       ■
                         M3.3 (SHAP) ■■→ M3.4 (Neg. sensitivity)■
                                        ■                         ■
                                        M3.7 (Gate 1→2) ■■■■■■■■■■
                                        ■                         ■
                                        M3.8 (Phase 1 report)    ■
                                        ■                         ■
                                ■■■■■■■■■■■■                        ■
                                ■                                 ■
                         M6.1 (CNN) ■■→ M6.2 (Filters)           ■
                          ■              ■                        ■
                         M6.3 (Saliency) M6.4 (Cell types)       ■
                          ■                   ■                   ■
                         M6.5 (Enh vs Prom)  ■                   ■
                          ■                   ■                   ■
                         M6.6 (Trade-off) ■■■■■■■                  ■
                          ■                                       ■
                         M6.7 (Gate 2→3)                          ■
                          ■                                       ■
                         M6.8 (Draft manuscript) ←■■■■■■■■■■■■■■■■■
                          ■
                   M12.1 (Pretrained) ■■→ M12.2 (Hybrid)
                                          ■
                                          M12.3 (Full manuscript)
                                          ■
                                    M12.4 (Model card)
                                    M12.5 (Code release)
                                    M12.6 (Submission)
```

Progress Tracking

This section should be updated as milestones are completed.

Milestone	Status	Date completed	Notes
M1.1	■ Not started	—	—
M1.2	■ Not started	—	—
M1.3	■ Not started	—	—
M1.4	■ Not started	—	—
M3.1	■ Not started	—	—
M3.2	■ Not started	—	—
M3.3	■ Not started	—	—
M3.4	■ Not started	—	—
M3.5	■ Not started	—	—
M3.6	■ Not started	—	—
M3.7	■ Not started	—	—
M3.8	■ Not started	—	—
M6.1	■ Not started	—	—
M6.2	■ Not started	—	—
M6.3	■ Not started	—	—
M6.4	■ Not started	—	—
M6.5	■ Not started	—	—
M6.6	■ Not started	—	—
M6.7	■ Not started	—	—
M6.8	■ Not started	—	—
M12.1	■ Not started	—	—
M12.2	■ Not started	—	—
M12.3	■ Not started	—	—
M12.4	■ Not started	—	—

M12.5	■ Not started	—	—
M12.6	■ Not started	—	—
M12.7	■ Not started	—	—
M12.8	■ Not started	—	—

End of Document 10 — Roadmap and Milestones
Next: 11 — Glossary & Project Memory